Comparative Molecular Virulence Typing and Antibiotic Resistance of Campylobacter Species at the Human–Animal–Environment Interface

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Abstract

This study holds significant importance due to its focus on *Campylobacter*, the leading bacterial cause of gastroenteritis worldwide, responsible for ∼96 million cases each year. By investigating the prevalence of both *Campylobacter jejuni* and *Campylobacter coli* in humans, animals, and the environment, this research sheds light on the broader impact of these pathogens, which can harm both human and animal populations. Traditional microbiological methods were implemented followed by optimized multiplex polymerase chain reaction targeting 16S rDNA and virulence gene markers by using specific primers. The findings revealed that a total of 219 *Campylobacter* isolates were recovered from 528 collected specimens from human, animal, and environmental sources. *Campylobacter* species showed a prevalence of 41.5%, with *C. jejuni* accounting for 53% and *C. coli* for 47%. Antimicrobial resistance rates were high, with tetracycline at 89%, ceftriaxone at 75%, cefotaxime at 70%, erythromycin at 69%, nalidixic acid at 54%, ciprofloxacin at 39%, and gentamicin at 23%. Commonly prevalent virulence-associated genes observed in the *Campylobacter* were *cad*F at 93%, *fla*A at 91%, *cdt*B at 88%, *che*Y at 86%, *sod*B at 78%, and *iam*A at 32%. The study confirmed multidrug-resistant *Campylobacter* prevalence at the human–animal–environment interface, harboring virulence-associated genes with potential harm to humans. Data analysis showed a nonsignificant ($p \geq 0.05$) correlation between virulence genes and antibiotic susceptibility. To effectively manage *Campylobacter* infections, a multifaceted strategy incorporating preventative interventions at different levels is required. This strategy should take into account practicability, effectiveness, and sustainability while strengthening surveillance systems and addressing the economics of disease prevention.